

White Paper: Direct PCR Detection for microbial pathogen.

Executive Summary

The cost and complexity of current nucleic acid isolation methods limits their use outside of the modern laboratory environment. To circumvent this limitation, BioPathogenix has developed two kits **BPX™ DNA Direct-Detect** and **BPX- Duo Direct-Detect** for the isolation of nucleic acid without complex and time-consuming extraction protocol. While **BPX™ DNA Direct-Detect targets diverse group of bacteria, fungus, parasites, and DNA viruses, BPX™ Duo Direct-Detect targets are formulated to detect gene of interest from wide range of bacteria and RNA viruses.** Both these reagent systems effectively release Nucleic acid for various downstream processing. The reagents used here are compatible with Reverse transcriptase (for cDNA synthesis) and DNA Polymerase and the enzymes and buffer system are specifically formulated to tolerate various inhibitors in different biological sample types. Due to its intrinsic simplicity and high sensitivity, BPX™ DNA Direct Detect and BPX™ Duo Direct Detect offers an excellent alternative to traditional nucleic acid extraction for multiplex real time PCR and various isothermal PCR.

Introduction

Polymerase Chain Reaction (PCR) is a fundamental tool in molecular biology technique widely used in research and diagnostic labs for the amplification of a specific DNA or RNA segment for detection and genotyping of various pathogens in biological samples. For the last two decades this technique has been a driving force

behind advances in the health care system. Traditionally to detect the desired Nucleic acid (DNA/RNA), preceding steps are necessary to isolate the nucleic acid from the given sample involving the lysis of the cells, purification, and subsequent recovery of nucleic acids from the cell lysate. These steps are tedious, time consuming, and not an affordable option for mass testing or in a low resource setting. At BioPathogenix we developed two reagent system; BPX™ DNA Direct Detect and BPX™ Duo direct detect for rapid and efficient preparation of nucleic acid from the various pathogen types. Formulation in BPX™ DNA Direct-Detect is compatible with DNA polymerase and target diverse group of bacteria (gram positive, gram negative, gram variable and PPLO group bacteria) fungus, parasites, and DNA viruses, whereas BPX™ Duo Direct-Detect is compatible with reverse transcriptase (for cDNA synthesis) and DNA polymerase and can release nucleic acid from various bacteria and viruses for multiplex real time PCR analysis. The strength of BPX™ Direct-Detect is its adaptability to a wide variety of application as demonstrated with diverse groups of human pathogens, multiple buffer system, diverse biological samples, and sensitivity for low copy number samples.

Here we provide an overview of BPX™ Direct Detect Kit, its advantages, challenges, applications, best practices, and emerging trends in the field. It serves as a resource for laboratory professionals, researchers, and stakeholders

interested in leveraging BPX™ Direct Detect for improved molecular testing.

Workflow and Methodology

BPX™ Direct Detect kits are based on a proprietary formulated reagent targeting distinct group of pathogens with different physical and chemical barriers releasing nucleic acid for multiplex qPCR or other downstream processing like isothermal PCR. BPX™ Direct-Detect bypasses multiple laborious sample processing steps together into a single step to minimize the time and effort traditionally required by other nucleic acid isolation method without compromising yield and reproducibility. Simplified overall workflow makes it more user-friendly, especially for laboratories with high sample volumes or those in a resource-limited settings. We have extensively analyzed their performances in terms robustness, sensitivity, and clinical relevance using large number of patient samples and contrived specimens with wide array of human pathogens in different sample matrixes. For the detailed workflow please refer to our user guide for BPX™ DNA Direct-Detect and BPX™ Duo Direct-Detect. A simplified workflow for the direct nucleic acid detection is below:

Mix sample with BPX™ Direct- Detect reagent.



Set up lysis at 55 °C - 8 min, 95 °C - 7 min.



Prepare the PCR reagents and set up PCR.

Direct Detection Vs. Nucleic Acid extraction.

Magnetic bead based or column based nucleic acid extraction is one of the widely used technologies for nucleic acid extraction. Several human pathogens representing diverse groups

were selected and tested for the efficiency and sensitivity of direct detection and compared with magnetic bead-based extraction. Multiplex real time PCR on these targets shows a strong correlation between the Ct values of extracted targets and BPX™ Direct Detect reagent, which makes this reagent a sensitive and cost-effective alternative for high throughput PCR for large scale diagnostics.

BPX™ Direct-Detect for diverse pathogens.

Microbial pathogens in clinical specimens are diverse and the recovery of nucleic acid is influenced by the physical and chemical properties of the pathogen. BPX™ Direct-Detect are formulated to ensure the release of nucleic acid from diverse group of pathogens having diverse physical barriers. Extensive analysis of its performances in terms of robustness, sensitivity, and clinical relevance using a wide number of human pathogens have shown it effectiveness in releasing nucleic acid from a wide range of pathogens. Table 1. summarizes different group of pathogens tested with BPX™ Direct Detect Kit.

Testing Human matrix for BPX™ Direct-Detect.

Nucleic acid detection from clinical samples is quite different from the cultured isolates of bacteria, fungi, or viruses. Each clinical specimen is unique, diverse and poses certain challenges for the release of nucleic acid. Therefor efficacy of BPX™ Direct Detect Kits were tested in various clinical samples. Samples for urine, wound and women's health panels were collected from diagnostic labs and analyzed with direct detection method and data were compared for direct detection and traditional extraction. Comparison of these data shows 100% concordance in the positive and negative targets tested across the panel. (Table 2,3,4)

Analytical Sensitivity (Limit of detection and Semiquant studies)

Limit of detection (LoD) is the lowest number of pathogens that can be detected at least in 95% of samples after nucleic acid is released by **BPX™ Direct-Detect Kit**. Three pathogens, *E.coli*, *S.aureus*, *C.albicans* representing gram-negative, gram-positive and fungal groups respectively were selected for this studies. Pathogens were diluted from 100000 cfu/μL to 1 cfu/μl by factor of 10 in each step. The determined LoD and their corresponding Ct values are summarized in Table 5.

These studies show our proprietary reagents for direct detections are fast powerful, and sensitive especially when we must handle a large sample volume. The reduction in sample processing time leads to faster results, making it particularly valuable in situations where rapid diagnosis is critical, such as infectious disease outbreaks.

Challenges and Considerations

While direct PCR offers numerous advantages, there are important considerations and challenges:

1. Inhibitors: Some sample types may contain substances that inhibit PCR reactions. However, at BioPathogenix we have done extensive studies on all possible inhibitors like Urea, creatinine IgG at a concentration above the biological level without any significant loss in the assay sensitivity.

3. Contamination Control: Although direct PCR reduces contamination risk, it does not eliminate it entirely. Laboratories must still implement stringent contamination control measures, such as separate workspaces and dedicated equipment.

Best Practices for Direct PCR Detection

1.Sample Preparation: Properly prepare and handle samples to minimize contamination and ensure optimal PCR conditions.

2.Optimization: Optimize PCR conditions, including primer design, cycling parameters, and reagent concentrations, for each specific target and sample type.

3.Validation: Validate direct PCR protocols for each application to ensure sensitivity and specificity.

4.Quality Control: Implement stringent quality control measures to monitor for contamination and maintain data integrity.

5.Contamination Mitigation: Use dedicated equipment, separate workspaces, and follow strict contamination control protocols.

6.Education and Training: Provide comprehensive training to laboratory personnel on the principles and techniques of direct detection of nucleic acid by real time PCR.

Conclusion

Direct PCR simplifies the overall workflow, making it more user-friendly for the lab personnels. This is especially beneficial for laboratories with high sample volumes or those in resource-limited settings. It offers significant advantages in terms of speed, simplicity, and cost-effectiveness. As technology continues to advance, the integration of direct PCR with emerging trends like point of care and NGS promises to further enhance its utility in molecular diagnostics, ultimately improving healthcare outcomes and scientific research.

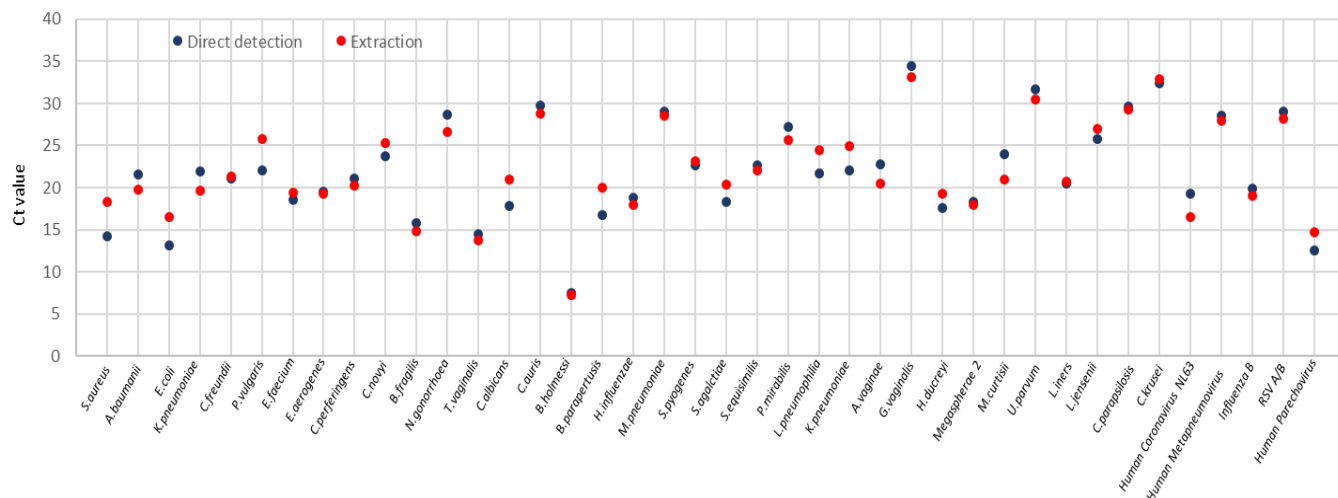


Figure 1. Comparison of BPX™ Direct-Detect with traditional extraction

gram -ve	gram -ve	gram +ve	protozoans
<i>Acinetobacter baumannii</i>	<i>Megasphaera 2</i>	<i>Atopobium vaginae</i>	<i>Entamoeba histolytica</i>
<i>Bacteroides fragilis</i>	<i>Moraxella catarrhalis</i>	<i>Clostridium difficile</i>	<i>Giardia lamblia</i>
<i>Bordetella holmesii</i>	<i>Morganella morganii</i>	<i>Clostridium novyi</i>	<i>Trichomonas vaginalis</i>
<i>Bordetella parapertussis</i>	<i>Neisseria gonorrhoeae</i>	<i>Clostridium perfringens</i>	Viruses
<i>Bordetella pertussis</i>	<i>Plesiomonas shigelloides</i>	<i>Clostridium septicum</i>	Herpes simplex virus (HSV1)
<i>Campylobacter</i>	<i>Prevotella bivia</i>	<i>Enterococcus faecalis</i>	Herpes simplex virus (HSV2)
<i>Chlamydia pneumoniae</i>	<i>Proteus mirabilis</i>	<i>Lactobacillus crispatus</i>	Influenza A
<i>Chlamydia trachomatis</i>	<i>Proteus vulgaris</i>	<i>Lactobacillus gasseri</i>	Influenza B
<i>Citrobacter freundii</i>	<i>Providencia stuartii</i>	<i>Lactobacillus iners</i>	Human Respiratory Syncytial Virus A
<i>Enterobacter aerogenes</i>	<i>Pseudomonas aeruginosa</i>	<i>Lactobacillus jensenii</i>	Human Respiratory Syncytial Virus B
<i>Enterobacter cloacae</i>	<i>Salmonella</i>	<i>Staphylococcus aureus</i>	Influenza A/H3
<i>Enterococcus faecium</i>	<i>Serratia marcescens</i>	<i>Staphylococcus saprophyticus</i>	Influenza A/H1-2009
<i>Escherichia coli</i>	<i>Vibrio cholerae</i>	<i>Streptococcus agalactiae</i>	Human Coronavirus OC43
<i>Haemophilus ducreyi</i>	<i>Yersinia enterocolitica</i>	<i>Streptococcus pneumoniae</i>	Human Coronavirus 229E
<i>Haemophilus influenzae</i>		<i>Streptococcus pyogenes</i>	Human Coronavirus NL63
<i>Haemophilus influenzae B</i>		Gram variable bacteria	Parainfluenza Virus 1
<i>Helicobacter pylori</i>		<i>Mobiluncus curtisii</i>	Parainfluenza Virus 2
<i>Kingella Kingae</i>		<i>Mobiluncus mulieris</i>	Parainfluenza Virus 3
<i>Klebsiella oxytoca</i>		PPLO Group	Parainfluenza Virus 4
<i>Klebsiella pneumoniae</i>		<i>Mycoplasma genitalium</i>	Human enterovirus
<i>Legionella longbeach</i>		<i>Mycoplasma hominis</i>	Human metapneumovirus A/B
<i>Legionella pneumophila</i>		<i>Mycoplasma pneumoniae</i>	Human parechovirus
		<i>Ureaplasma urealyticum</i>	Human adenovirus B3

Table 1. Diverse human pathogen tested and identified with BPX™ Direct- Detect Kit

Collecto tube		Saureus	A.baumannii	E.coli	K.pneumoniae	P.aeruginosa	K.oxitoca	E.coliae	P.mirabilis	M.morganii	P.aeruginosa	C.freundii	E.aerogenes	E.aerogenes	M.hominis	U.urealyticum	P.stuartii	S.marcescens	S.saprophyticus	S.saprophyticus	Callicoccus	C.graminis	C.graminis	C.graminis	C.graminis	C.graminis	C.graminis	C.graminis
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Table 2. Urine sample analyzed with BPX™ Direct-Detect Kit. (Grey cells are the results from traditional extraction and white cells are the results from direct detection) *(continued..)*

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	Sample ID	Saureus	Abamarii	Efaecis	Ecoli	K.pneumoniae	P.aeruginosa	K.oxytoca	E.coliae	P.mirabilis	M.morgani	P.vulgaris	C.frendii	Eaerogens	Efaecium	M.hominis	U.urealyticum	P.stuarti	S.marcescens	S.saprophyticus	Sagalidae	Callicens	C.glabrata	C.krusei	C.tropicalis	C.auris	C.lustanae	C.oprapsilosis
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Table 2. Urine sample analyzed with BPX™ Direct-Detect Kit (Grey cells are the results from traditional extraction and white cells are the results from direct detection). (continued..)

	Collectio tube		S.aureus	A.baumannii	E.faecalis	E.coli	K.pneumoniae	P.aeruginosa	K.oxytoca	E.cloacae	P.mirabilis	M.morgani	P.vulgaris	C.freundii	Eaerogens	E.faecium	M.hominis	U.urealyticum	P.stuartii	S.marcescens	S.saprophyticus	S.agalctiae	C.albicans	C.glabrata	C.kruzei	C.tropicalis	C.auris	C.lusitaniae	C.parapsilosis	
BD Vacutainer -UA preservative tube	Sample ID																													
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Table 2. Urine sample analyzed with BPX™ Direct-Detect Kit (Grey cells are the results from traditional extraction and white cells are the results from direct detection). (continued..)

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Table 3. Women's health panel analyzed with BPX™ Direct-Detect Kit (Grey cells are the results from traditional extraction and white cells are the results from direct detection).

Collectio tube	Sample ID	S.aureus	A.baumannii	E.faecalis	E.coli	K.pneumoniae	P.vulgaris	C.freundii	M.morgai	P.mirabilis	S.pyogenes	E.aerogenes	B.fragilis	E.faecium	K.kingae	C.novyi	C.septicum	C.perferingens
BPX -Amies media	Sample 1								+	+								
	Sample 1								+	+								
	Sample 2									+								
	Sample 2									+								
	Sample 3	+																
	Sample 3	+																
	Sample 4	+																
	Sample 4	+																
	Sample 5	+																
	Sample 5	+																
	Sample 6	+						+		+								
	Sample 6	+						+		+								
	Sample 7			+	+	+			+				+					
	Sample 7			+	+	+			+				+					
	Sample 8	+	+															
	Sample 8	+	+															
	Sample 9	+		+		+												
	Sample 9	+		+		+												
	Sample 10	+		+		+												
	Sample10	+		+		+												

Table 4. Wound panel analyzed with BPX™ Direct Detect Kit (Grey cells are the results from traditional extraction and white cells are the results from direct detection).

cfu/ μ L	<i>E.coli</i>	<i>S.aureus</i>	<i>C.albicans</i>
100000	15.665 $\pm$ 0.019	18.88 $\pm$ 0.052	18.656 $\pm$ 0.086
10000	18.791 $\pm$ 0.116	21.699 $\pm$ 0.124	21.187 $\pm$ 0.053
1000	22.488 $\pm$ 0.415	24.306 $\pm$ 0.154	25.313 $\pm$ 0.067
100	25.896 $\pm$ 0.057	26.261 $\pm$ 0.112	28.571 $\pm$ 0.231
10	29.046 $\pm$ 0.013	28.351 $\pm$ 0.15	31.136 $\pm$ 0.738
1	32.374 $\pm$ 0.211	30.401 $\pm$ 0.419	33.608 $\pm$ 0.153

Table 5. Limit of detection (LoD) for representative pathogens with BPX™ Direct Detect Kit.